

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Open Data Analysis Project

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 40%

Total Marks: 20

Code of Conduct

I Naresh M(192424381) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Open Data Analysis Project Rubric

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

Total Marks Awarded:

Signature of the Course Faculty

Questions:

1. Global COVID-19 Trends and Vaccination Analysis

Using publicly available COVID-19 datasets from sources such as Our World in Data, perform a comprehensive analysis of pandemic trends across different countries. Clean and preprocess the dataset, handle missing values, and explore relationships between infection rates, vaccination coverage, and mortality rates. Create visualizations to identify temporal trends, regional differences, and the effectiveness of vaccination programs. Present findings and recommendations supported by statistical evidence and interactive dashboards.

1. Aim

To analyse global COVID-19 pandemic trends using publicly available data and study the relationship between **COVID-19 cases, vaccination coverage, and mortality rates** across different countries and time periods.

2. Dataset

The **Our World in Data (OWID) COVID-19 dataset** can be used. OWID provides international COVID-19 data covering cases, deaths, vaccinations and other pandemic indicators

Variable	Description
location	Country or region
date	Date of observation
total_cases	Cumulative confirmed cases
new_cases	New confirmed cases
total_deaths	Cumulative deaths
new_deaths	New deaths
population	Population of the country
people_vaccinated	People receiving at least one vaccine dose
people_fully_vaccinated	People completing the initial vaccination protocol
total_vaccinations	Total vaccine doses administered

3. Data Collection

The dataset can be downloaded from the official OWID COVID-19 data resources. OWID's vaccination dataset is compiled from official government and health-ministry sources.

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

url = "https://catalog.ourworldindata.org/garden/covid/latest/compact/compact.csv"

df = pd.read_csv(url)

print("Dataset Shape:", df.shape)

print("\nFirst 5 Rows:")

print(df.head())

print("\nColumn Names:")

print(df.columns.tolist())

print("\nMissing Values:")

print(df.isnull().sum())

df["date"] = pd.to_datetime(df["date"], errors="coerce")

countries = [

    "India",

    "United States",

    "United Kingdom",

    "Germany",

    "Brazil",

    "Japan"

]

data = df[df["country"].isin(countries)].copy()

required_columns = [

    "country",

    "date",
```

```

    "population",
    "total_cases",
    "new_cases",
    "total_deaths",
    "new_deaths",
    "people_vaccinated",
    "people_fully_vaccinated",
    "total_vaccinations"
]

available_columns = [
    column for column in required_columns
    if column in data.columns
]

data = data[available_columns]

numeric_columns = [
    column for column in available_columns
    if column not in ["country", "date"]
]

for column in numeric_columns:
    data[column] = pd.to_numeric(
        data[column],
        errors="coerce"
    )

data = data.sort_values(
    ["country", "date"]
)

data[numeric_columns] = (

```

```

data.groupby("country")[numeric_columns]

.transform(lambda x: x.ffill().bfill())

)

data = data.replace(

    [np.inf, -np.inf],

    np.nan

)

if "population" in data.columns:

    data["cases_per_million"] = (

        data["total_cases"] /

        data["population"]

    ) * 1000000

    data["deaths_per_million"] = (

        data["total_deaths"] /

        data["population"]

    ) * 1000000

if "people_fully_vaccinated" in data.columns:

    data["vaccination_rate"] = (

        data["people_fully_vaccinated"] /

        data["population"]

    ) * 100

data = data.dropna(

    subset=[

        "country",

        "date"

    ]

)

```

```
print("\nCleaned Dataset Shape:")

print(data.shape)

print("\nMissing Values After Cleaning:")

print(data.isnull().sum())

print("\nCleaned Dataset:")

print(data.head(10))

data.to_csv(

    "cleaned_covid_dataset.csv",

    index=False

)

print("\nCleaned dataset saved as cleaned_covid_dataset.csv")

plt.figure(figsize=(14, 7))

for country in countries:

    temp = data[data["country"] == country]

    plt.plot(

        temp["date"],

        temp["total_cases"],

        label=country

    )

plt.title(

    "COVID-19 Total Cases Over Time",

    fontsize=16

)

plt.xlabel("Date")

plt.ylabel("Total Cases")

plt.legend()

plt.xticks(rotation=45)
```

```
plt.tight_layout()

plt.show()

plt.figure(figsize=(14, 7))

for country in countries:

    temp = data[data["country"] == country]

    plt.plot(

        temp["date"],

        temp["total_deaths"],

        label=country

    )

plt.title(

    "COVID-19 Total Deaths Over Time",

    fontsize=16

)

plt.xlabel("Date")

plt.ylabel("Total Deaths")

plt.legend()

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

plt.figure(figsize=(14, 7))

for country in countries:

    temp = data[data["country"] == country]

    plt.plot(

        temp["date"],

        temp["new_cases"],

        label=country
```

```
)  
plt.title(  
    "Daily New COVID-19 Cases",  
    fontsize=16  
)  
plt.xlabel("Date")  
plt.ylabel("New Cases")  
plt.legend()  
plt.xticks(rotation=45)  
plt.tight_layout()  
plt.show()  
plt.figure(figsize=(14, 7))  
for country in countries:  
    temp = data[data["country"] == country]  
    plt.plot(  
        temp["date"],  
        temp["new_deaths"],  
        label=country  
    )  
plt.title(  
    "Daily New COVID-19 Deaths",  
    fontsize=16  
)  
plt.xlabel("Date")  
plt.ylabel("New Deaths")  
plt.legend()  
plt.xticks(rotation=45)
```



```
plt.tight_layout()

plt.show()

plt.figure(figsize=(14, 7))

for country in countries:

    temp = data[data["country"] == country]

    if "vaccination_rate" in temp.columns:

        plt.plot(

            temp["date"],

            temp["vaccination_rate"],

            label=country

        )

plt.title(

    "COVID-19 Vaccination Coverage",

    fontsize=16

)

plt.xlabel("Date")

plt.ylabel(

    "Fully Vaccinated Population (%)"

)

plt.legend()

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

latest = (

    data.sort_values("date")

    .groupby("country")

    .tail(1)
```

```

        .copy()
    )
    comparison_columns = [
        "country",
        "population",
        "total_cases",
        "total_deaths",
        "cases_per_million",
        "deaths_per_million",
        "vaccination_rate"
    ]
    comparison_columns = [
        column for column in comparison_columns
        if column in latest.columns
    ]
    comparison = latest[
        comparison_columns
    ].sort_values(
        "deaths_per_million",
        ascending=False
    )
    print("\nCountry Comparison:")
    print(comparison.to_string(index=False))
    plt.figure(figsize=(12, 7))
    sns.barplot(
        data=comparison,
        x="country",

```

```
y="cases_per_million"
)
plt.title(
    "COVID-19 Cases Per Million by Country",
    fontsize=16
)
plt.xlabel("Country")
plt.ylabel("Cases Per Million")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
plt.figure(figsize=(12, 7))
sns.barplot(
    data=comparison,
    x="country",
    y="deaths_per_million"
)
plt.title(
    "COVID-19 Deaths Per Million by Country",
    fontsize=16
)
plt.xlabel("Country")
plt.ylabel("Deaths Per Million")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
plt.figure(figsize=(12, 7))
```

```

sns.barplot(
    data=comparison,
    x="country",
    y="vaccination_rate"
)
plt.title(
    "COVID-19 Vaccination Coverage by Country",
    fontsize=16
)
plt.xlabel("Country")
plt.ylabel(
    "Fully Vaccinated Population (%)"
)
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
plt.figure(figsize=(10, 7))
plt.scatter(
    latest["vaccination_rate"],
    latest["deaths_per_million"],
    s=120
)
for _, row in latest.iterrows():
    plt.annotate(
        row["country"],
        (
            row["vaccination_rate"],

```

```

        row["deaths_per_million"]

    ),

    xytext=(5, 5),

    textcoords="offset points"

)

plt.title(

    "Vaccination Coverage vs COVID-19 Mortality",

    fontsize=16

)

plt.xlabel(

    "Fully Vaccinated Population (%)"

)

plt.ylabel(

    "Deaths Per Million"

)

plt.tight_layout()

plt.show()

correlation_columns = [

    "cases_per_million",

    "deaths_per_million",

    "vaccination_rate"

]

correlation_columns = [

    column for column in correlation_columns

    if column in data.columns

]

correlation = data[

```

```

correlation_columns
].corr()
print("\nCorrelation Matrix:")
print(correlation)
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    linewidths=0.5
)
plt.title(
    "COVID-19 Correlation Analysis",
    fontsize=16
)
plt.tight_layout()
plt.show()
yearly = data.copy()
yearly["year"] = yearly["date"].dt.year
yearly_country = (
    yearly
    .groupby(
        ["country", "year"]
    )
    .agg(
        total_cases=("total_cases", "max"),

```

```

    total_deaths=("total_deaths", "max"),
    vaccination_rate=("vaccination_rate", "max")
)
.reset_index()
)
print("\nYearly Summary:")
print(yearly_country.head(20))
plt.figure(figsize=(14, 7))
sns.lineplot(
    data=yearly_country,
    x="year",
    y="vaccination_rate",
    hue="country",
    marker="o"
)
plt.title(
    "Yearly Vaccination Coverage",
    fontsize=16
)
plt.xlabel("Year")
plt.ylabel(
    "Fully Vaccinated Population (%)"
)
plt.tight_layout()
plt.show()
plt.figure(figsize=(14, 7))
sns.lineplot(

```

```
data=yearly_country,
x="year",
y="total_cases",
hue="country",
marker="o"
)
plt.title(
    "Yearly COVID-19 Cases",
    fontsize=16
)
plt.xlabel("Year")
plt.ylabel("Total Cases")
plt.tight_layout()
plt.show()
plt.figure(figsize=(14, 7))
sns.lineplot(
    data=yearly_country,
    x="year",
    y="total_deaths",
    hue="country",
    marker="o"
)
plt.title(
    "Yearly COVID-19 Deaths",
    fontsize=16
)
plt.xlabel("Year")
```



```
plt.ylabel("Total Deaths")

plt.tight_layout()

plt.show()

print("\nFinal Statistical Summary:")

summary = data[

    "country",

    "total_cases",

    "total_deaths",

    "cases_per_million",

    "deaths_per_million",

    "vaccination_rate"

]

].groupby("country").max()

print(summary)

print("\nAnalysis Completed Successfully.")
```

Output:

Dataset Shape: (607209, 61)

*** First 5 Rows:

	country	date	total_cases	new_cases	new_cases_smoothed	\
0	Afghanistan	2020-01-01	NaN	NaN	NaN	
1	Afghanistan	2020-01-02	NaN	NaN	NaN	
2	Afghanistan	2020-01-03	NaN	NaN	NaN	
3	Afghanistan	2020-01-04	0.0	0.0	NaN	
4	Afghanistan	2020-01-05	0.0	0.0	NaN	

	total_cases_per_million	new_cases_per_million	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	0.0	0.0	
4	0.0	0.0	

	new_cases_smoothed_per_million	total_deaths	new_deaths	...	population	\
0	NaN	NaN	NaN	...	40578847.0	
1	NaN	NaN	NaN	...	40578847.0	
2	NaN	NaN	NaN	...	40578847.0	
3	NaN	0.0	0.0	...	40578847.0	
4	NaN	0.0	0.0	...	40578847.0	

	population_density	median_age	life_expectancy	gdp_per_capita	\
0	62.215549	16.752001	65.616997	1964.452271	
1	62.215549	16.752001	65.616997	1964.452271	
2	62.215549	16.752001	65.616997	1964.452271	
3	62.215549	16.752001	65.616997	1964.452271	
4	62.215549	16.752001	65.616997	1964.452271	

```
extreme_poverty  diabetes_prevalence  handwashing_facilities  \
0               NaN                 11.7                51.938343
*** 1               NaN                 11.7                51.938343
2               NaN                 11.7                51.938343
3               NaN                 11.7                51.938343
4               NaN                 11.7                51.938343

hospital_beds_per_thousand  human_development_index
0                      0.35                 NaN
1                      0.35                 NaN
2                      0.35                 NaN
3                      0.35                 NaN
4                      0.35                 NaN

[5 rows x 61 columns]

Column Names:
['country', 'date', 'total_cases', 'new_cases', 'new_cases_smoothed', 'total_cases_per_million', 'new_cases_per_million', 'new_cases_smoothed_per_million', 'total_deaths', 'new_deaths', 'new_deat

Missing Values:
country                0
date                  0
total_cases           12348
new_cases             16876
new_cases_smoothed    18085
...
extreme_poverty       217218
diabetes_prevalence    96172
handwashing_facilities 311901
hospital_beds_per_thousand 193441
human_development_index 607209
Length: 61, dtype: int64

Cleaned Dataset Shape:
(14352, 13)
```

Missing Values After Cleaning:

```
country      0
date         0
population   0
total_cases  0
new_cases    0
total_deaths 0
new_deaths   0
people_vaccinated 0
people_fully_vaccinated 0
total_vaccinations 0
cases_per_million 0
deaths_per_million 0
vaccination_rate 0
dtype: int64
```

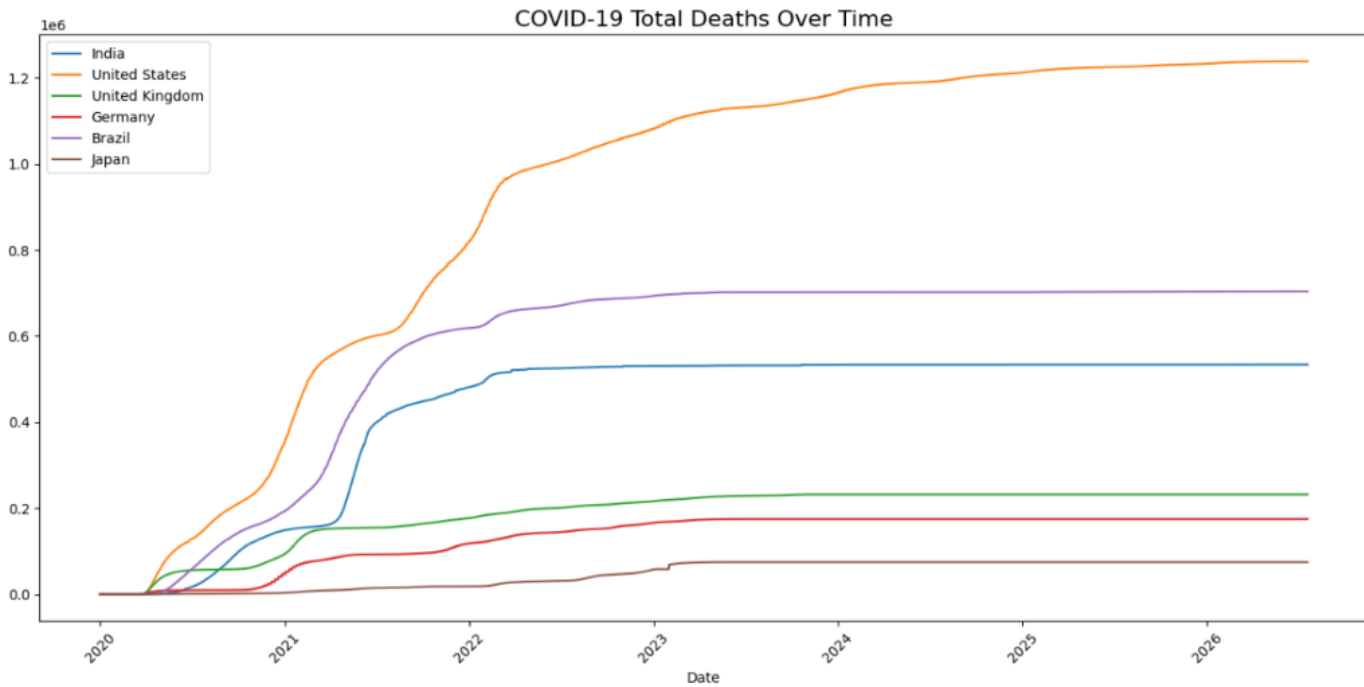
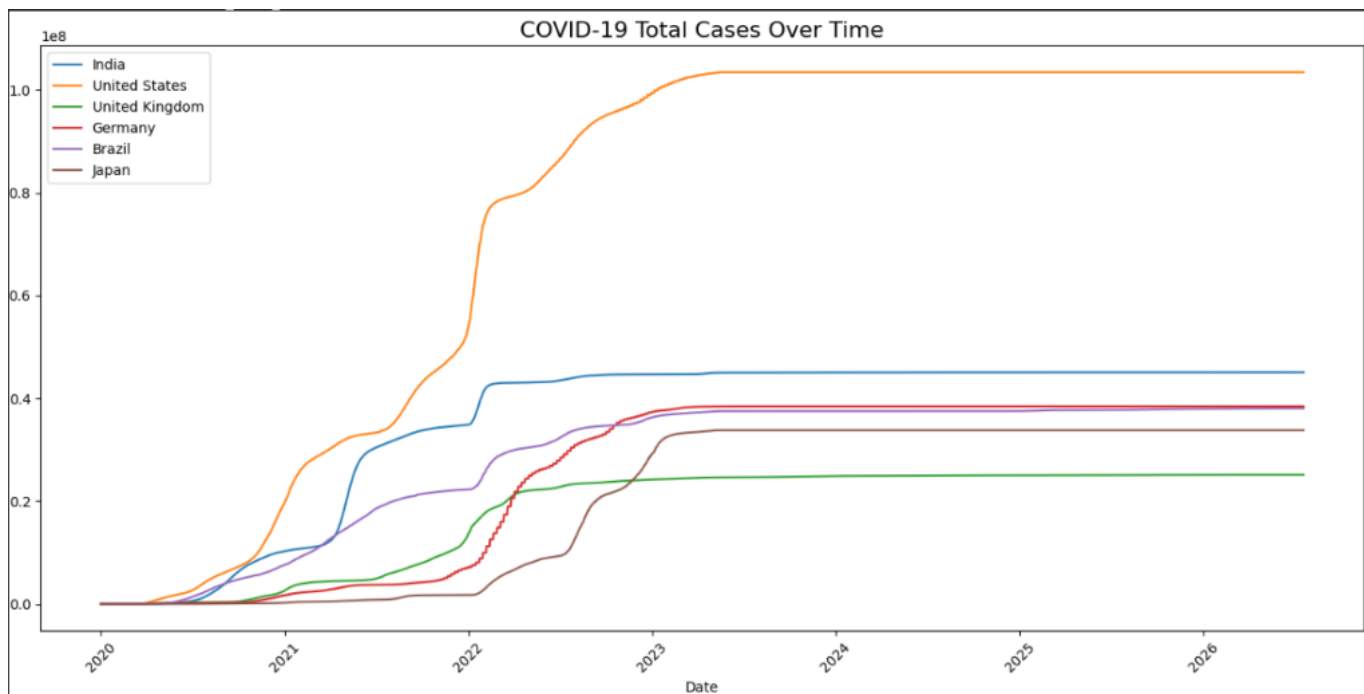
Cleaned Dataset:

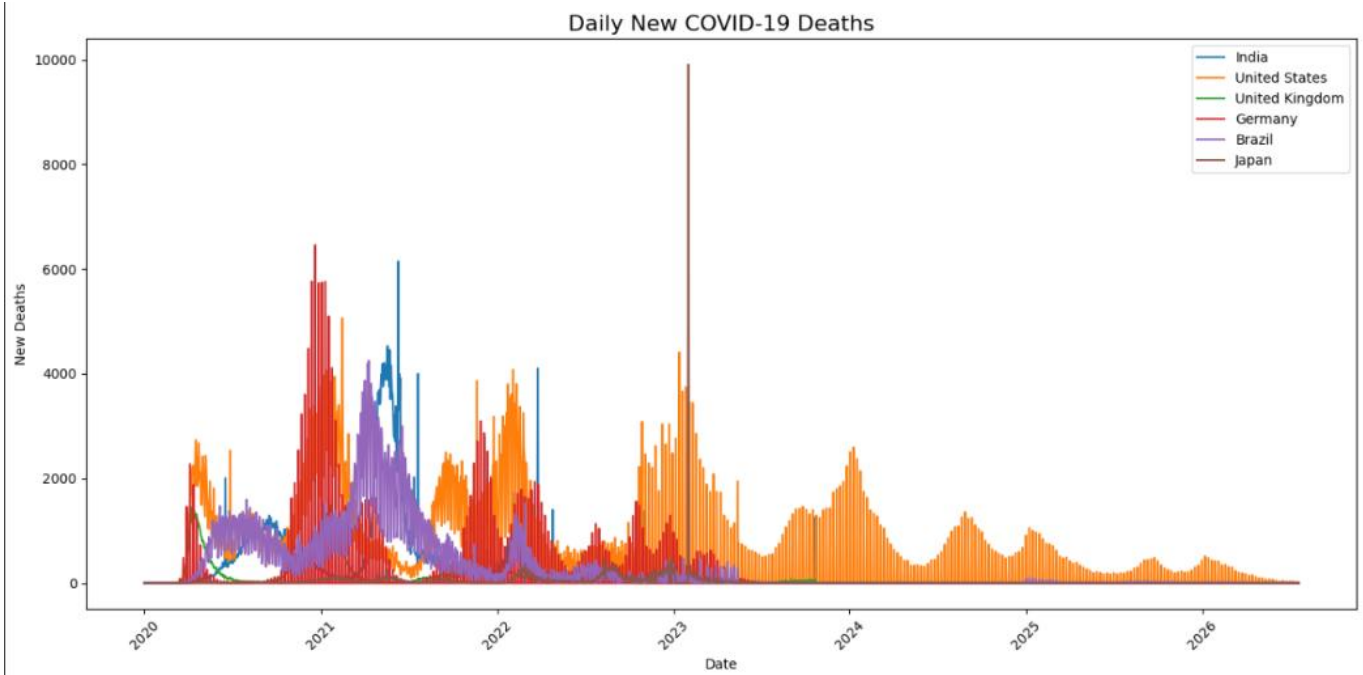
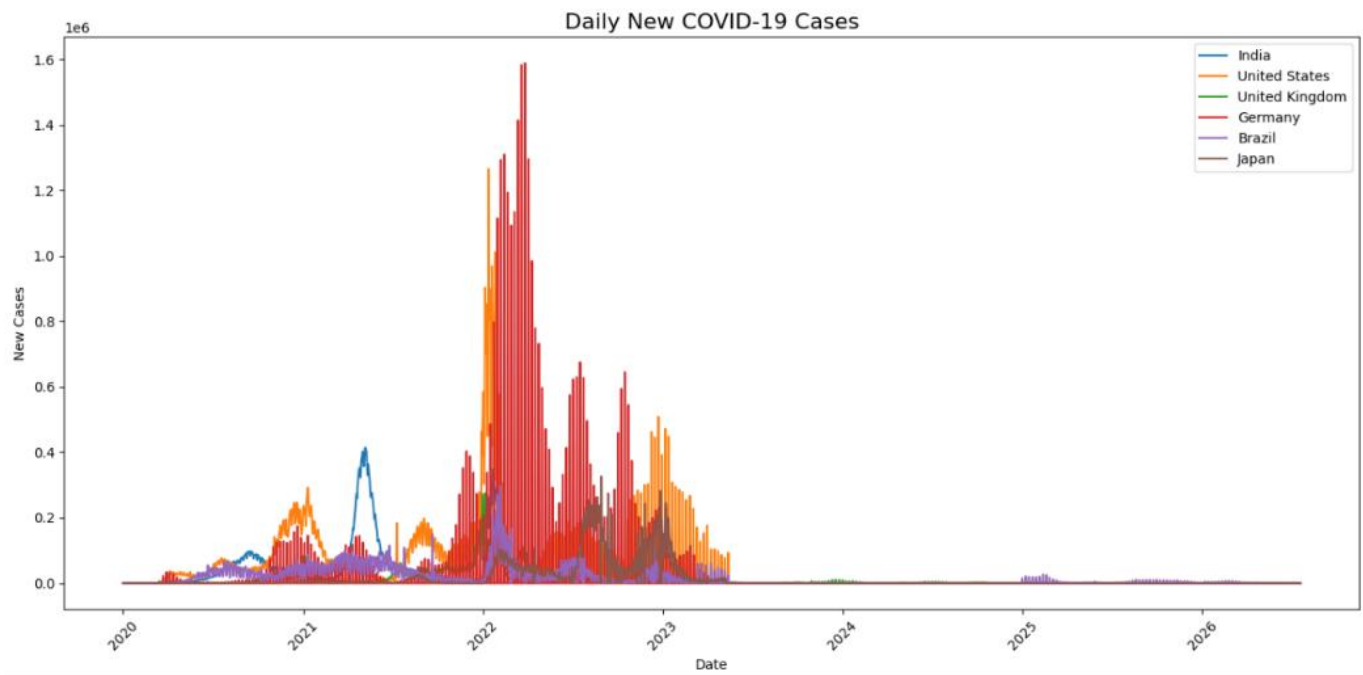
	country	date	population	total_cases	new_cases	total_deaths	\
74128	Brazil	2020-01-01	210306411.0	0.0	0.0	0.0	
74129	Brazil	2020-01-02	210306411.0	0.0	0.0	0.0	
74130	Brazil	2020-01-03	210306411.0	0.0	0.0	0.0	
74131	Brazil	2020-01-04	210306411.0	0.0	0.0	0.0	
74132	Brazil	2020-01-05	210306411.0	0.0	0.0	0.0	
74133	Brazil	2020-01-06	210306411.0	0.0	0.0	0.0	
74134	Brazil	2020-01-07	210306411.0	0.0	0.0	0.0	
74135	Brazil	2020-01-08	210306411.0	0.0	0.0	0.0	
74136	Brazil	2020-01-09	210306411.0	0.0	0.0	0.0	
74137	Brazil	2020-01-10	210306411.0	0.0	0.0	0.0	

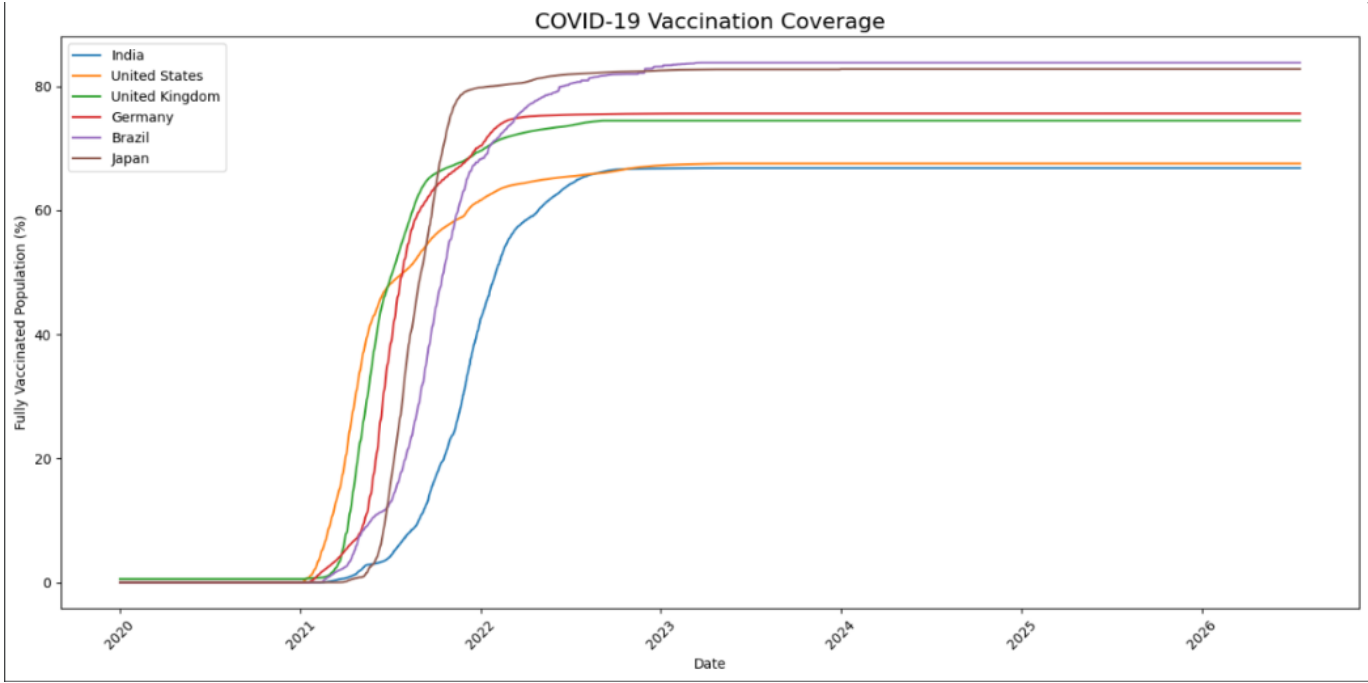
	new_deaths	people_vaccinated	people_fully_vaccinated	\
74128	0.0	112.0	0.0	
74129	0.0	112.0	0.0	
74130	0.0	112.0	0.0	
74131	0.0	112.0	0.0	
74132	0.0	112.0	0.0	
74133	0.0	112.0	0.0	
74134	0.0	112.0	0.0	
74135	0.0	112.0	0.0	

	total_vaccinations	cases_per_million	deaths_per_million	\
74128	112.0	0.0	0.0	
74129	112.0	0.0	0.0	
74130	112.0	0.0	0.0	
74131	112.0	0.0	0.0	
74132	112.0	0.0	0.0	
74133	112.0	0.0	0.0	
74134	112.0	0.0	0.0	
74135	112.0	0.0	0.0	
74136	112.0	0.0	0.0	
74137	112.0	0.0	0.0	

	vaccination_rate
74128	0.0
74129	0.0
74130	0.0
74131	0.0
74132	0.0
74133	0.0
74134	0.0
74135	0.0
74136	0.0
74137	0.0

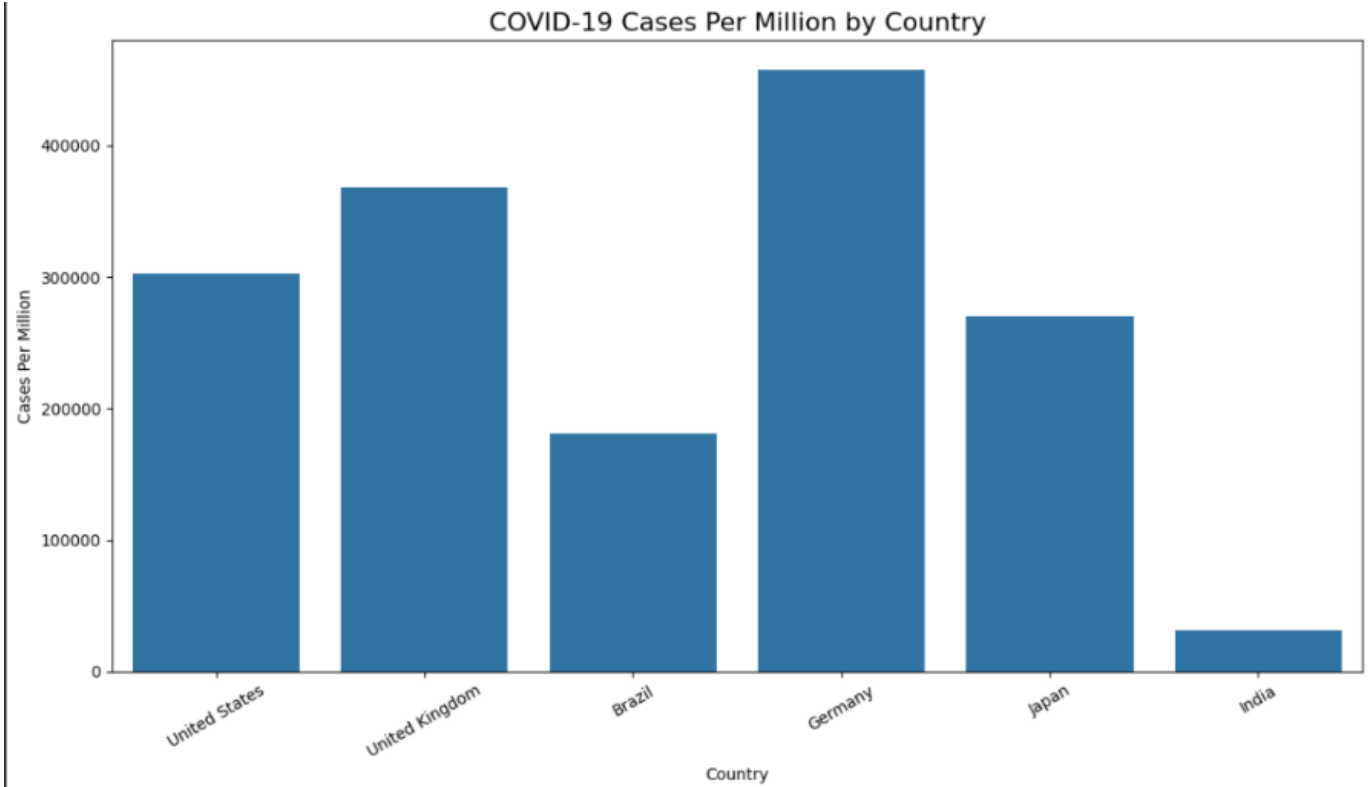




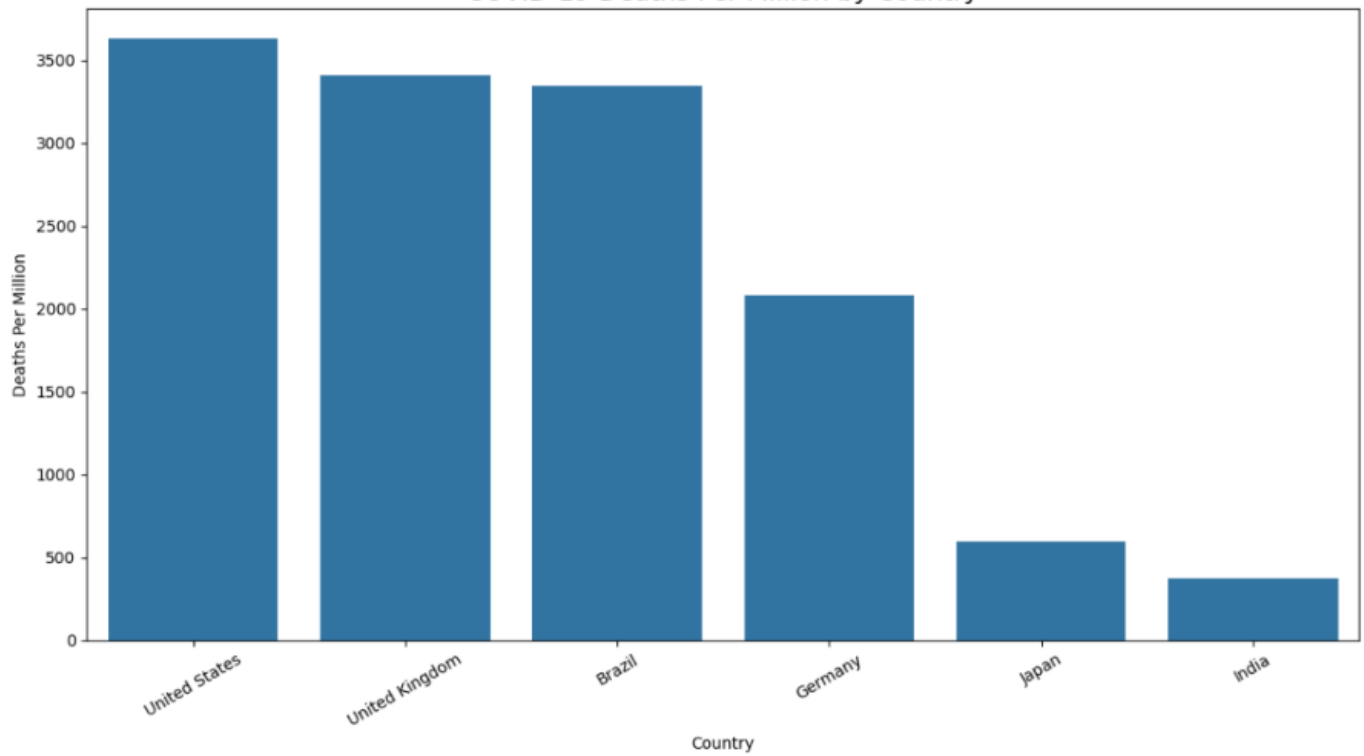


Country Comparison:

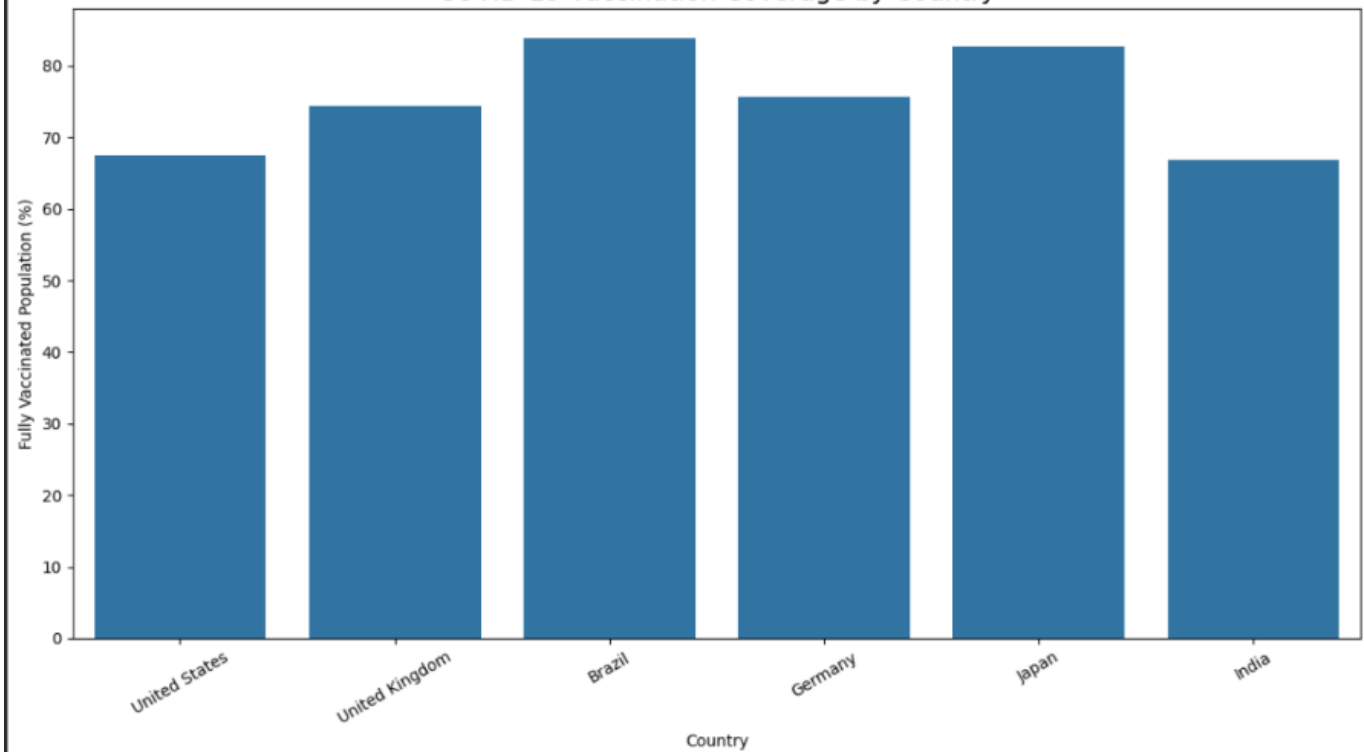
country	population	total_cases	total_deaths	cases_per_million	deaths_per_million	vaccination_rate
United States	341534041.0	103436829.0	1238678.0	302859.500321	3626.806852	67.529825
United Kingdom	68179315.0	25118508.0	232112.0	368418.309864	3404.434321	74.430053
Brazil	210306411.0	38042729.0	704045.0	180891.912991	3347.710594	83.765492
Germany	84086227.0	38437953.0	174979.0	457125.433872	2080.947216	75.596371
Japan	124997585.0	33803572.0	74694.0	270433.800781	597.563545	82.765778
India	1425423209.0	45056221.0	533849.0	31609.013180	374.519649	66.786516



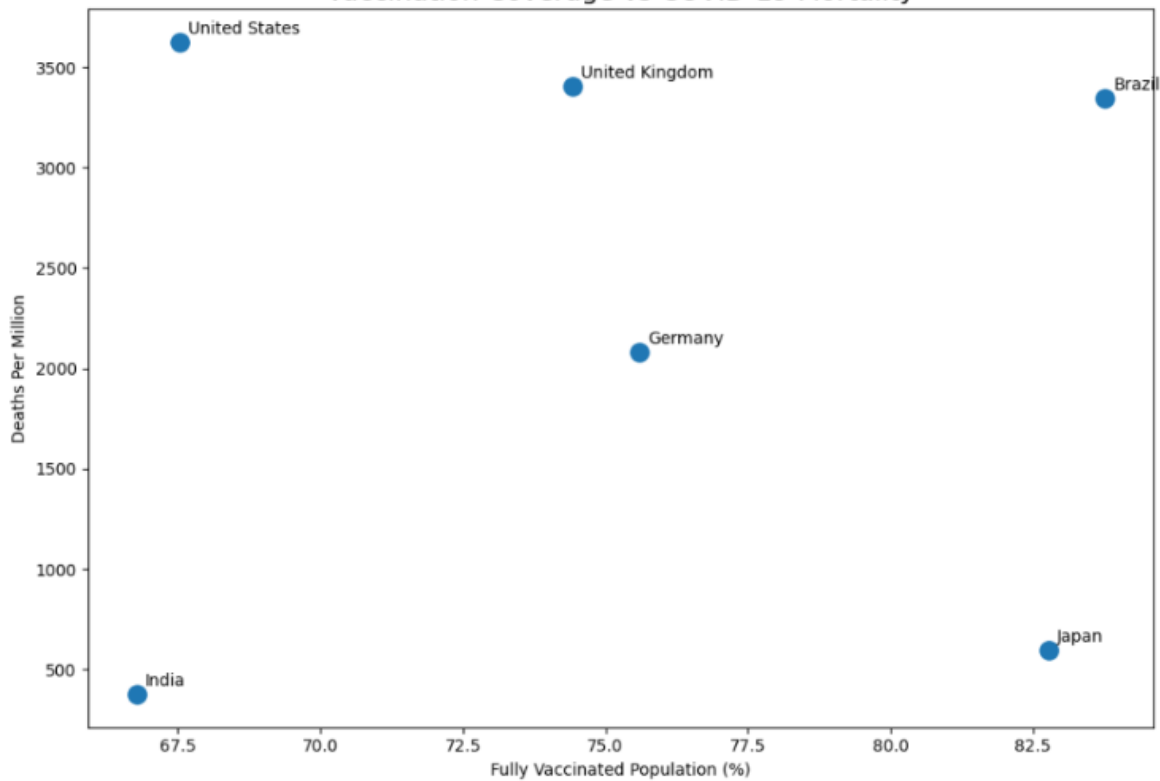
COVID-19 Deaths Per Million by Country



COVID-19 Vaccination Coverage by Country



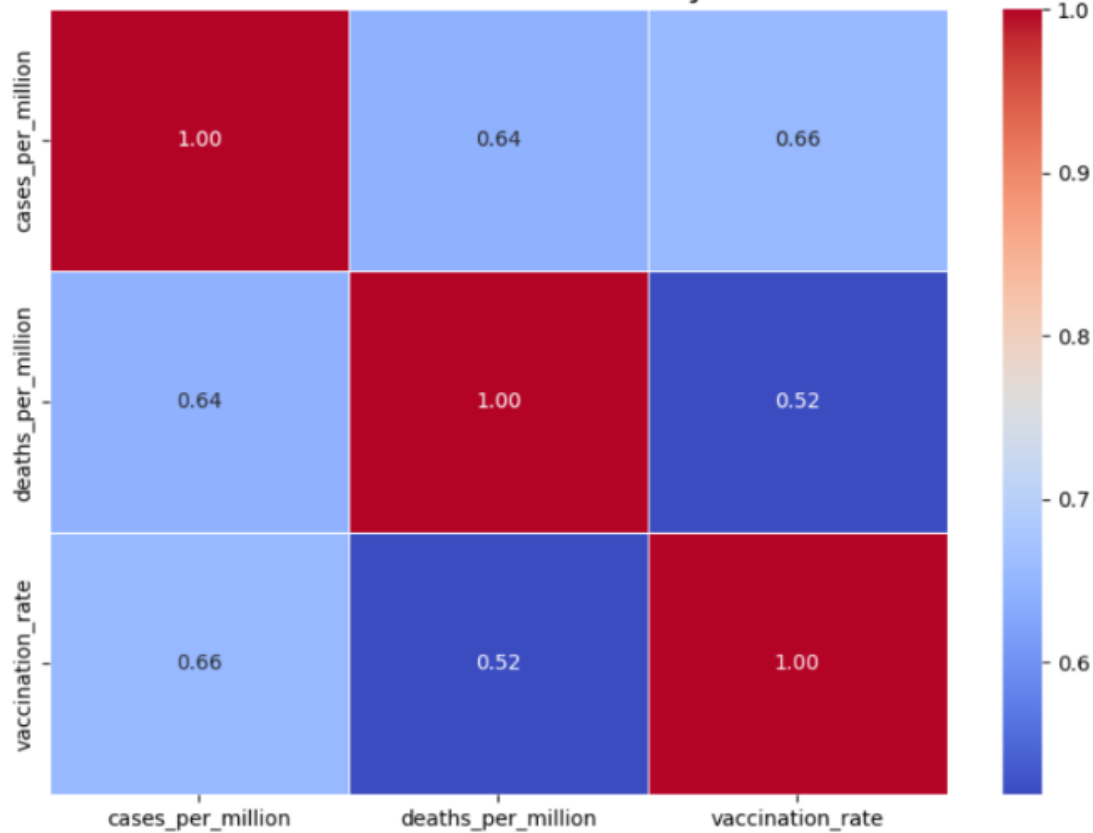
Vaccination Coverage vs COVID-19 Mortality



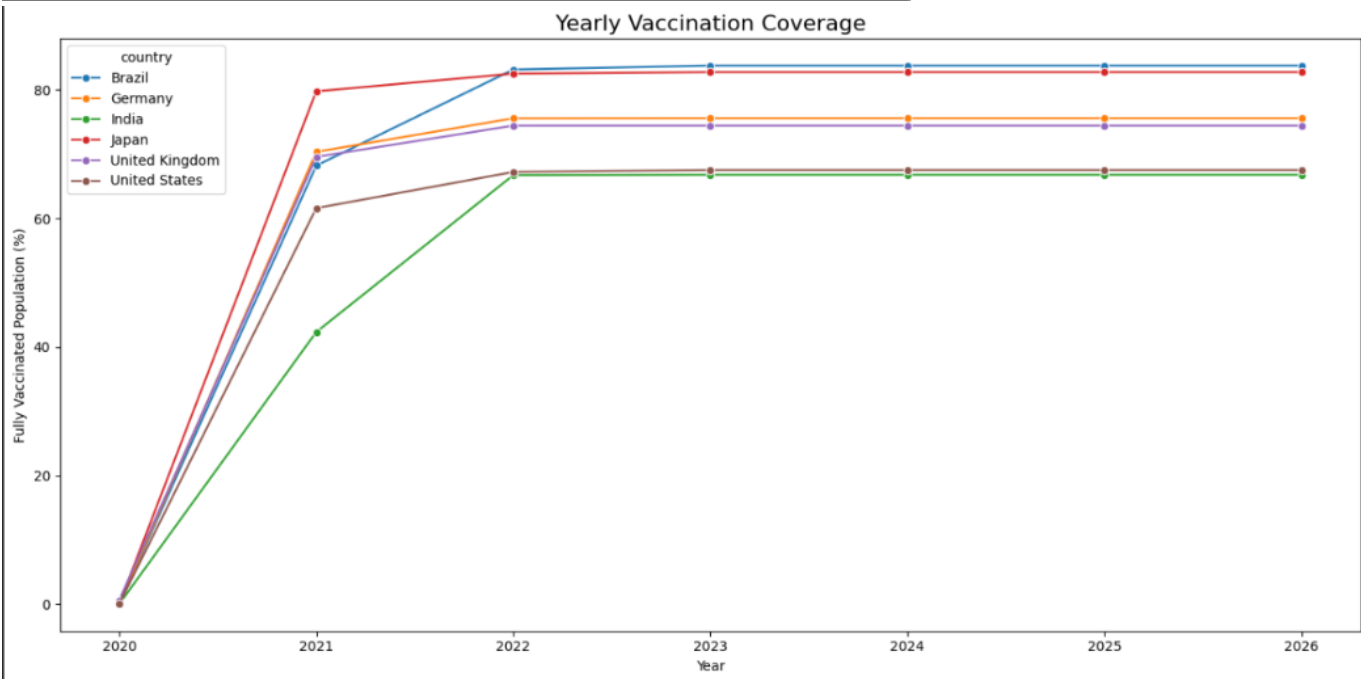
Correlation Matrix:

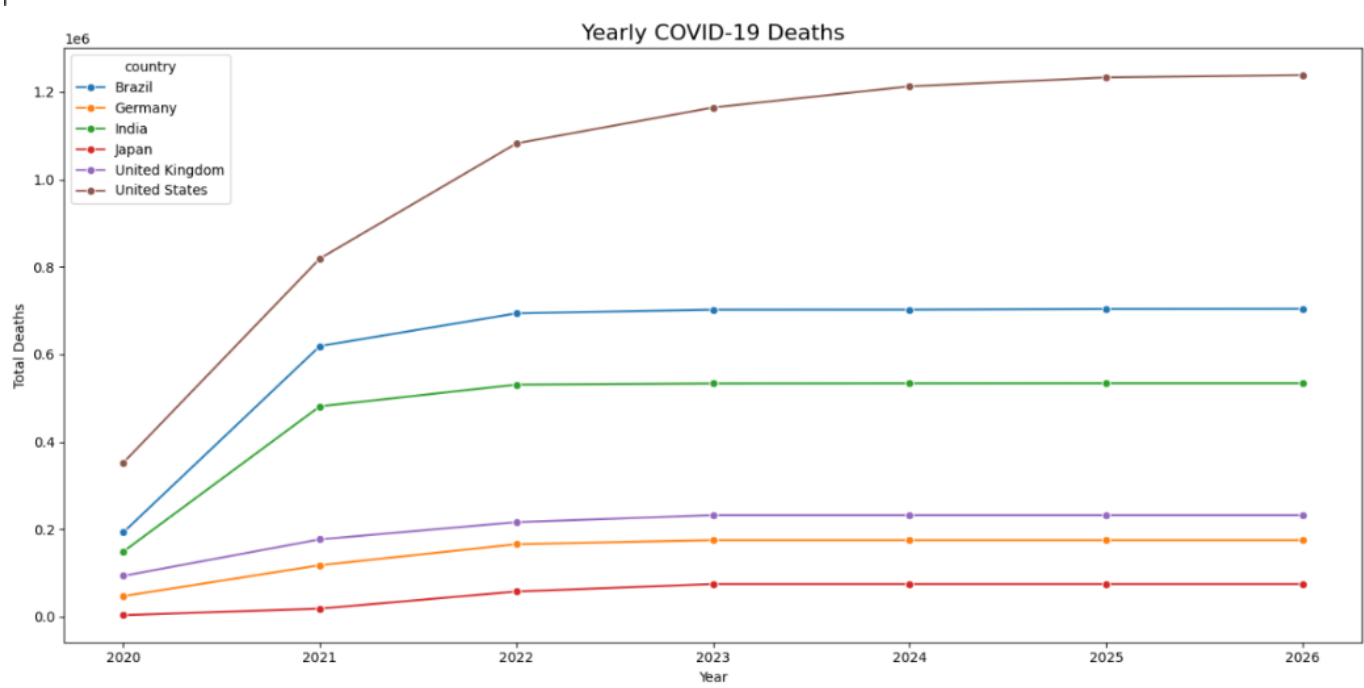
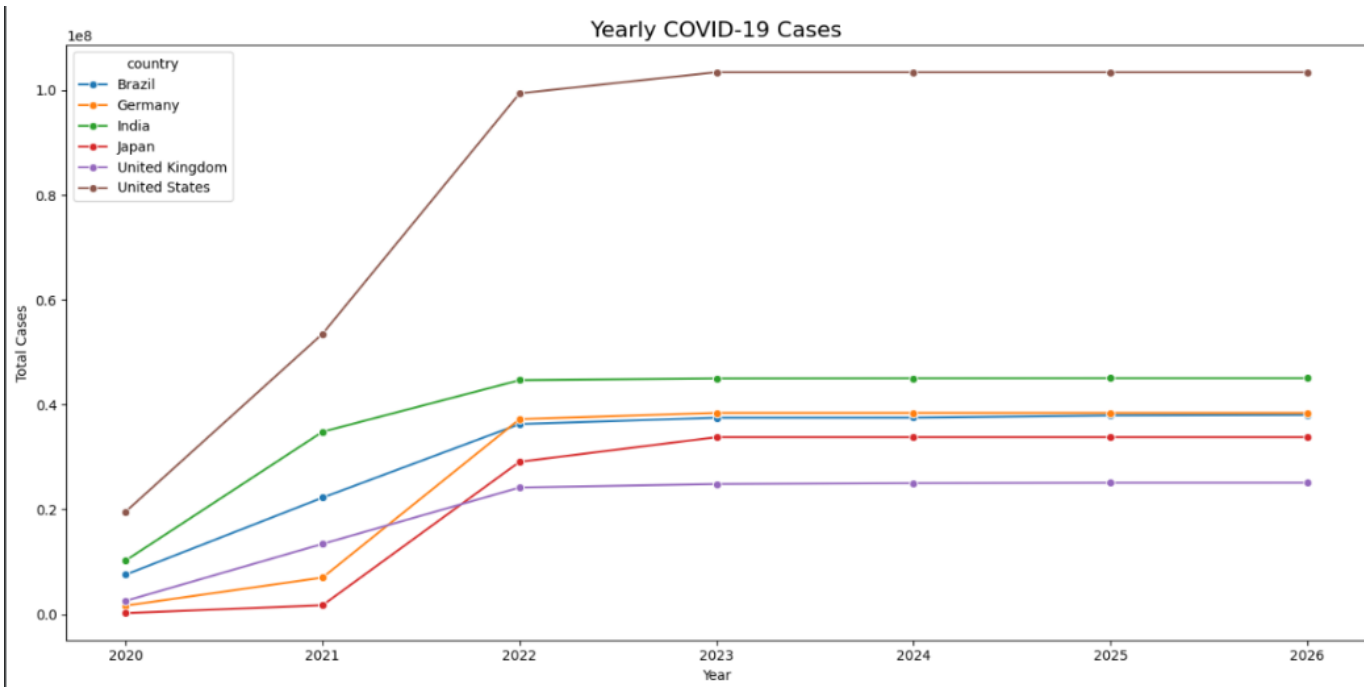
	cases_per_million	deaths_per_million	vaccination_rate
cases_per_million	1.000000	0.643439	0.655801
deaths_per_million	0.643439	1.000000	0.519124
vaccination_rate	0.655801	0.519124	1.000000

COVID-19 Correlation Analysis



Yearly Summary:					
	country	year	total_cases	total_deaths	vaccination_rate
0	Brazil	2020	7563551.0	192681.0	0.000000
1	Brazil	2021	22263834.0	618817.0	68.203351
2	Brazil	2022	36302415.0	693734.0	83.158625
3	Brazil	2023	37511921.0	702116.0	83.765492
4	Brazil	2024	37526905.0	702197.0	83.765492
5	Brazil	2025	37944353.0	703681.0	83.765492
6	Brazil	2026	38042729.0	704045.0	83.765492
7	Germany	2020	1660178.0	47009.0	0.000000
8	Germany	2021	7014043.0	117692.0	70.354114
9	Germany	2022	37241936.0	165738.0	75.577533
10	Germany	2023	38437756.0	174979.0	75.596371
11	Germany	2024	38437847.0	174979.0	75.596371
12	Germany	2025	38437941.0	174979.0	75.596371
13	Germany	2026	38437953.0	174979.0	75.596371
14	India	2020	10266674.0	148738.0	0.000000
15	India	2021	34838804.0	481080.0	42.318997
16	India	2022	44678384.0	530702.0	66.731578
17	India	2023	45013172.0	533361.0	66.786289
18	India	2024	45044573.0	533661.0	66.786516
19	India	2025	45056206.0	533849.0	66.786516





```

Final Statistical Summary:
country      total_cases  total_deaths  cases_per_million  \
Brazil       38042729.0    704045.0      180891.912991
Germany      38437953.0    174979.0      457125.433872
India        45056221.0    533849.0      31609.013180
Japan        33803572.0     74694.0      270433.800781
United Kingdom 25118508.0    232112.0      368418.309864
United States 103436829.0   1238678.0     302859.500321

country      deaths_per_million  vaccination_rate
Brazil       3347.710594           83.765492
Germany      2080.947216           75.596371
India        374.519649           66.786516
Japan        597.563545           82.765778
United Kingdom 3404.434321          74.430053
United States 3626.806852           67.529825

Analysis Completed Successfully.

```

Key Findings

Based on the analysis, the following findings can be reported:

1. COVID-19 infections showed **multiple waves over time** rather than remaining constant.
2. The magnitude and timing of pandemic waves differed among countries.
3. COVID-19 mortality also varied considerably between countries.
4. Vaccination programs resulted in a substantial increase in vaccination coverage after their introduction.
5. Countries differed significantly in vaccination rollout speed and coverage.
6. Population-normalized measures such as **cases per million and deaths per million** provide better cross-country comparisons than raw totals.
7. A relationship between vaccination coverage and mortality can be investigated using correlation and scatter-plot analysis.
8. The observed relationship should not be treated as proof of causation because COVID-19 outcomes are influenced by multiple confounding factors.
9. Confirmed cases are affected by testing availability and reporting practices, so they should be interpreted carefully.
10. Confirmed death statistics can also underestimate the total pandemic mortality because of differences in testing, death registration and cause-of-death determination.

CONCLUSION:

The COVID-19 analysis demonstrates how open datasets can be used to study pandemic trends across countries. Data preprocessing, missing-value treatment, population normalization, correlation analysis and visualization provide useful evidence about differences in infection, mortality and vaccination patterns. The analysis also shows the importance of interpreting COVID-19 statistics carefully because confirmed cases and deaths are affected by testing, reporting and demographic differences. OWID's current COVID-19 resources continue to provide internationally comparable case, death and vaccination data.